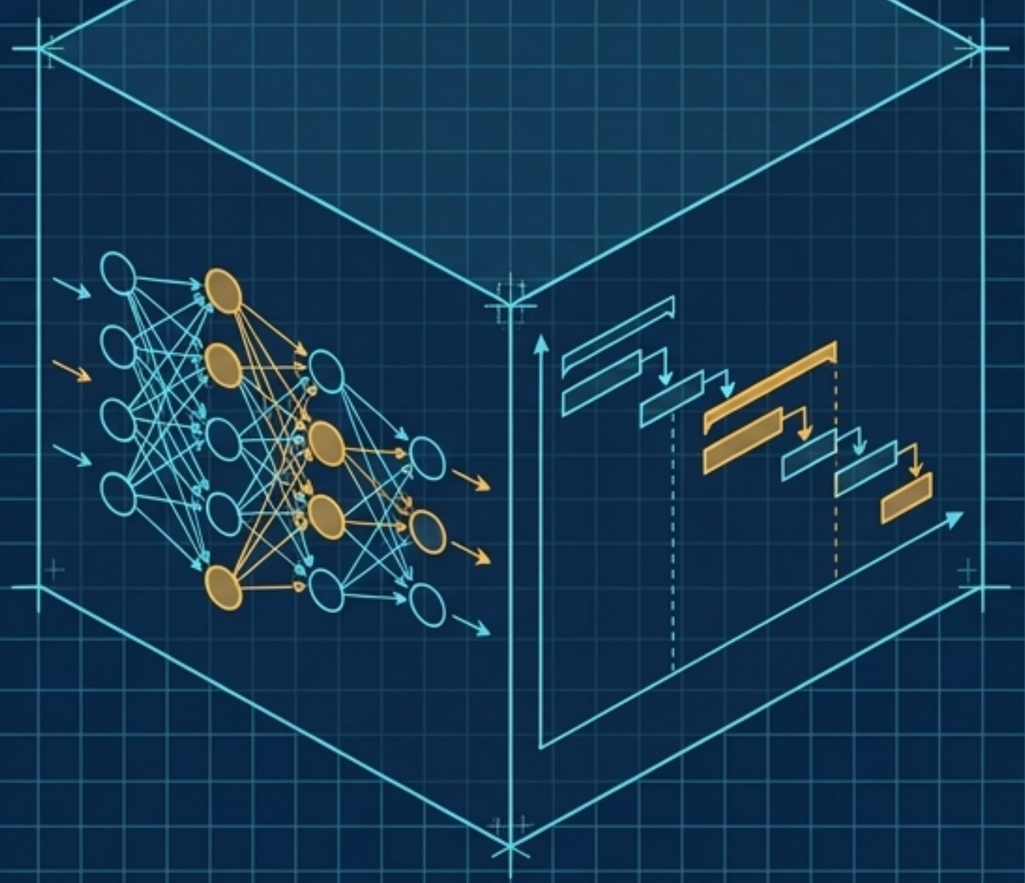
FIRST REVIEW

AI-Enhanced Metaheuristics Control for Project Scheduling Using Neural Networks

Subtitle: An Intelligent Decision-Making Framework for Resource-Constrained Project Scheduling Optimization



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Evaluation Phase: First Project Review (Review 1)

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Presentation Outline



01. Abstract: Core problem definition and 8-point research scope

02. Introduction: RCPSP complexity, NP-hardness & AI conductor paradigm

03. Literature Survey: Evaluation of prior research benchmarks and gaps

04. Problem Identification: Goal, combinatorial barrier, and tooling gap

05. Proposed System & Workflow: Architecture, 8 enhancements, and end-to-end pipeline

06. Review 1 Status (30% Complete): Implementation of SSGS + 4 baseline algorithms

07. Program Outputs & Verification: Execution logs for GA, SA, BCO, and PRH

08. Conclusion & References: Deliverables, Phase 2 roadmap, and citations

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Abstract

Every project is made up of many small tasks that depend on each other and share limited resources such as workers, machines, or money, and finding the best order and timing to complete all tasks as quickly as possible is known as **Project Scheduling**, a problem that becomes extremely difficult to solve exactly as the project grows larger. To handle this, researchers use smart search techniques called **metaheuristic algorithms**, such as **Genetic Algorithm (GA)**, **Simulated Annealing (SA)**, **Bee Colony Optimization (BCO)**, and simple **Priority Rule basic based Heuristics (PRH)**, but since no single algorithm works best for every project, a recent study trained a small **Artificial Neural Network** that looks at three simple numbers describing a project's structure and instantly recommends which of **these four algorithms is likely to give the best schedule**, avoiding the need to run every algorithm and compare results.

This project builds on that idea and improves it in eight ways, with the most significant contribution being the **addition of two brand-new algorithms** that the original study did not include: **Tabu Search (TS)**, which remembers recently tried solutions and avoids repeating them so the search does not get stuck going back and forth between the same options, and **Particle Swarm Optimization (PSO)**, inspired by how a flock of birds searches for food together, where each candidate schedule adjusts itself based on its own best result and the best result found by the whole group to quickly converge on strong schedules; together these expand the neural network's choices from four algorithms to six, giving it a wider and more capable set of strategies to recommend from.

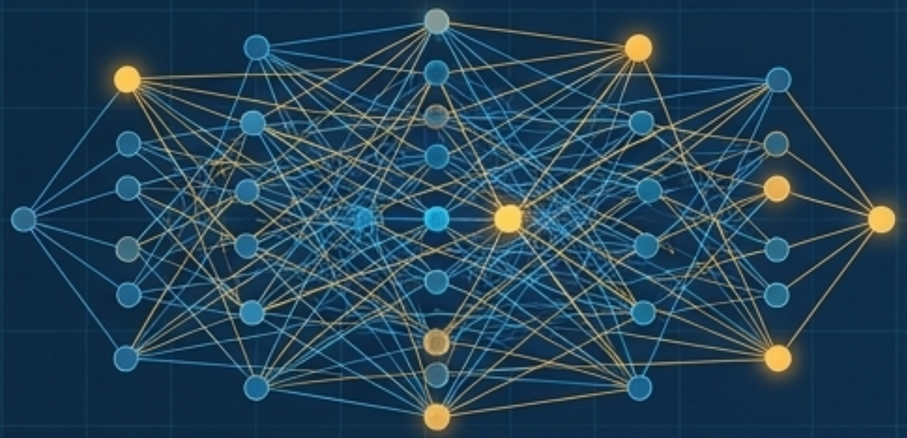
The remaining improvements include using richer project features for better accuracy, predicting the actual expected schedule duration rather than only naming an algorithm, providing a confidence score with each recommendation, correcting a data imbalance issue so rarely-best algorithms are still recognized reliably, testing the system on newly generated unseen projects to check real-world generalization, using a **Graph Neural Network** to read a project's task structure directly, and a self-adjusting controller that can switch algorithms mid-search if the current one stalls, together producing a faster, more accurate, and adaptable assistant that saves computing effort while giving schedules as good as or better than the original system.

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Introduction: RCPSP Complexity, NP-Hardness & AI Project Scope

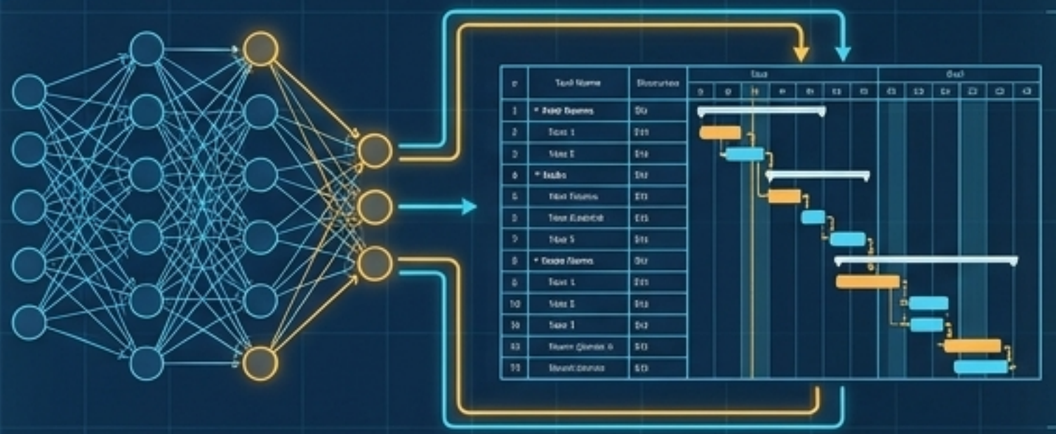
This project addresses the challenge of efficient project scheduling by leveraging Artificial Intelligence. The goal is to automate the selection of the most effective algorithm for any given project, moving beyond manual trial-and-error. We focus on the Resource-Constrained Project Scheduling Problem (RCPSP), a complex task where the objective is to minimize total project duration (makespan) under strict resource and dependency constraints.

The Problem & Complexity (RCPSP & NP-Hard)



- **The Challenge (RCPSP):** Requires minimizing total project duration (makespan) while respecting strict resource limits and task dependencies.
- **The NP-Hard Definition:** RCPSP is classified as NP-hard. This means that as the number of tasks and constraints grows, the computational time to find the exact optimal schedule increases exponentially, making exact solutions infeasible for large-scale, real-world projects.
- **The Implication:** Traditional exact methods fail for large instances, necessitating the use of metaheuristic algorithms and AI for finding sufficiently good, near-optimal solutions efficiently.

The Solution & Project Scope



- **Project Scope:** To develop an AI-driven system that automates the selection of the best-suited metaheuristic algorithm for a given RCPSP instance, optimizing for minimum makespan.
- **The AI Approach:** Leveraging a neural network to analyze a project's unique topological features and predict the most effective metaheuristic algorithm, acting as an 'Intelligent Conductor'.
- **Automated Selection Benefit:** Guarantees a high-quality solution far more efficiently than manual trial-and-error, bypassing the computational intractability of exact methods.

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Literature Survey: Evaluating Prior Approaches

Sl. No.	Paper	Authors	Year	Methodology	Advantages	Disadvantages
1	Quantum-Inspired Genetic Algorithm for Resource-Constrained Project Scheduling	H. M. H. Saad et al.	2021	Uses a genetic algorithm (evolutionary search) improved with quantum computing ideas for better exploration.	Proposed a Quantum-Inspired Algorithm (QIGA) using quantum operators.	Focuses on improving a single algorithm; expensive; no choice of which algorithm is best.
2	Learning the Quality of Machine Permutations in Job Shop Scheduling	A. Corsini et al.	2022	Trains a neural network to predict the quality of different machine schedules.	Formulated Job Shop Scheduling using neural networking.	Limited to Job Shop Scheduling; requires large training and multiple algorithms.
3	APULSE: A Scalable Hybrid Algorithm for the RCSPP on Large-Scale Dense Graphs	L. Chen et al.	2026	Combines multiple search techniques into a hybrid algorithm for large-scale scheduling problems.	Introduced APULSE, a hybrid algorithm for dense graphs.	Designed for shortest path, not general RCSPP; fixed for all instances.
4	Exploiting Ontology Recommendation Using Text Categorization Approach	A. M. Al-Zoubi et al.	2020	Uses text categorization to recommend relevant knowledge structures (ontologies).	Proposed text categorization for ontologies.	Focused on ontology, not scheduling; limited scalability.
5	Ontology-Based Personalized Course Recommendation Framework	M. A. Al-Bakry et al.	2019	Uses a structured knowledge base (ontology) to recommend personalized courses.	Developed an ontology-based recommendation algorithm.	Application-specific to education; does not handle resource constraints.

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Problem Statement

Project managers struggle to optimize schedules for large projects because current tools cannot handle the complex challenge of balancing limited resources.



The Goal: Minimize total project time while strictly following resource limits and task order.

The Barrier: As projects grow, finding the best schedule becomes computationally too complex for perfect solutions.

The Tooling Gap: Existing tools are too basic, forcing managers to use manual trial-and-error instead of smart, automated solutions.

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Current Baseline System vs. Proposed Enhanced System

Current Baseline System:

- **Limited Toolkit**: Restricted to only 4 basic algorithms (GA, SA, BCO, PRH).
- **Rigid Execution**: Relying on manual trial-and-error across candidate heuristics.
- **Basic Inputs**: Relies on shallow, simplistic feature extraction parameters.

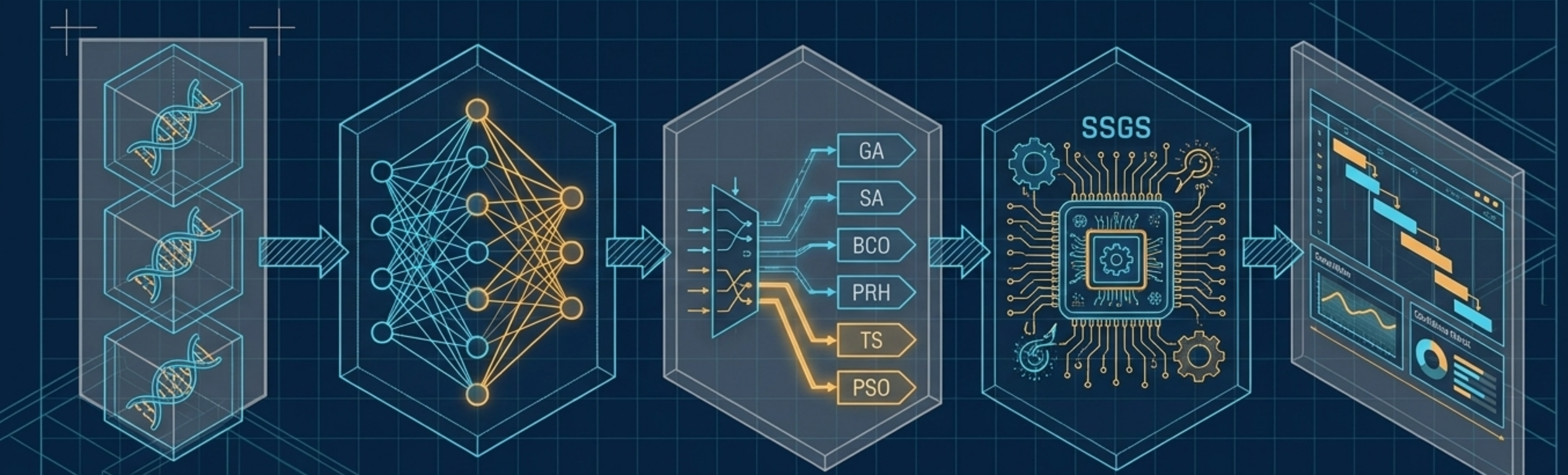


Proposed Enhanced System (8 Key Improvements Over Baseline):

- **Bigger Toolbox** (6 Tools): Like having a diverse toolkit, including 'memory' (Tabu Search to avoid repeating mistakes) and 'teamwork' (PSO to find solutions faster) for any project.
- **Better Project Check-up**: Uses advanced scans to fully understand a project's structure, like a doctor getting a detailed MRI before treatment.
- **Clear, Usable Answers**: Gives a realistic completion time and a confidence score, like a GPS predicting arrival time and traffic.
- **Strong & Flexible**: Learns from rare cases, works on real projects, and even switches strategies mid-job, like a self-adjusting thermostat.

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Proposed System Workflow: The Intelligent Conductor



Stage 1 — Project Input Features (The 'DNA'): Network Complexity, Resource Factor, Resource Strength.

Stage 2 — Graphical Neural Network (The Brain): Instantly analyses the topology to predict the optimal optimal strategy.

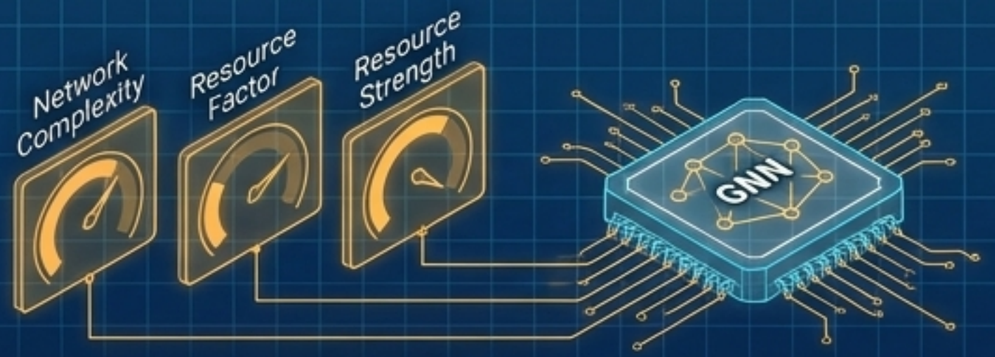
Stage 3 — Algorithm Selection (The Toolkit): Routes the project to the single best algorithm out of six (GA, SA, BCO, PRH, TS, PSO).

Stage 4 — Serial Scheduling Generation Scheme (SSGS) (Execution): Constructs the physical schedule under strict constraints.

Stage 5 — Optimal Project Makespan (Output): Delivers the fastest possible completion time & confidence metrics.

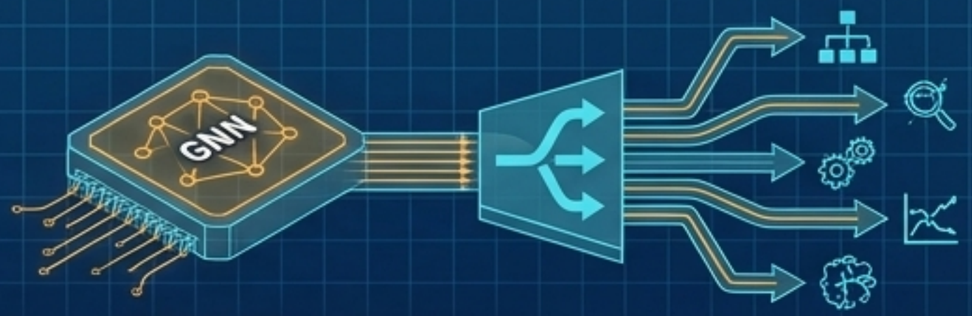
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Workflow: Stage 1 to Stage 5 Technical Breakdown



Stage 1: Topological Extraction

Evaluates project network descriptors without brute-force tests: Network Complexity, Resource Factor, and Resource Strength, augmented with Graph Neural Network (GNN) task DAG ingestion.



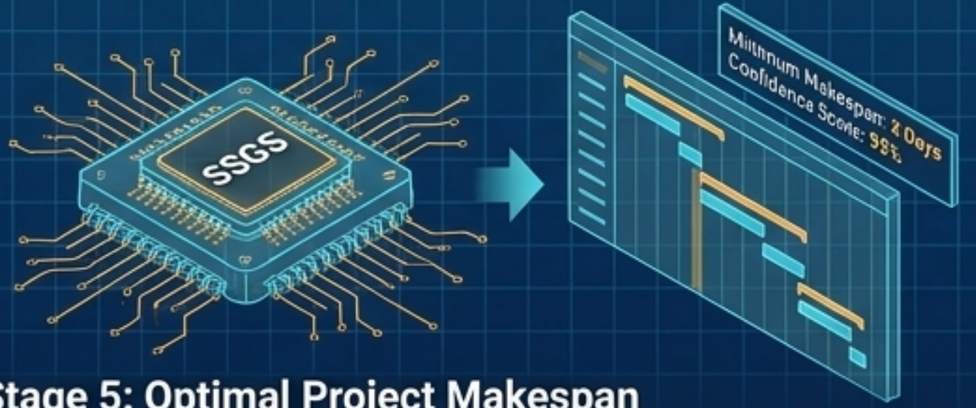
Stage 2 & 3: Selection Engine

Routes the project to the single optimal algorithm out of six: GA & BCO (population & swarm searches), SA & PRH (trajectory & construction heuristics), **NEW** - Tabu Search (TS), **NEW** - Particle Swarm Optimization (PSO).



Stage 4: SSGS Schedule Construction

Selected metaheuristic employs a Serial Schedule Generation Scheme (SSGS) using random key priority lists, monitored by a Self-Adjusting Controller.



Stage 5: Optimal Project Makespan

Delivers minimum makespan, outputs predictive duration prior to run, and attaches a statistical confidence score.

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Review 1 Implementation Status: 30% Completed (Review 1 Milestone)



Phase 2 Roadmap (Remaining 70% Scope):

- Algorithm suite expansion: Tabu Search (TS) and Particle Swarm Optimization (PSO).
- Feature extraction pipeline and PyTorch Geometric GNN DAG ingestion.
- Multi-task prediction head (makespan duration regression + confidence scoring).
- Self-adjusting dynamic switching controller for mid-search stagnation.

Delivered Baseline Modules (30% Total Scope Completed):

- Serial Schedule Generation Scheme (SSGS): Discrete timeline tracking, topological predecessor verification, and renewable resource conflict detection.
- Algorithm 1 — Genetic Algorithm (GA): Activity permutation chromosome, Order Crossover (OX), tournament selection, antitological repair mutation.
- Algorithm 1 — Genetic Algorithm (iiA): Order Crossover (OX), tourmament selection, elitism, and topological repair mutation.
- Algorithm 2 — Simulated Annealing (SA): Adjacent precedence-feasible swap neighborhood generator, geometric cooling, and Metropolis acceptance neighborhood generator, geometric cooling, and Metropolis acceptance criterion.
- Algorithm 3 — Bee Colony Optimization (BCO): Employed bee exploitation, onlooker bee selection, and scout bee random re-initialization.
- Algorithm 4 — Priority Rule-based Heuristics (PRH): Deterministic priority rules (Latest Finish Time, Most Successors, Minimum Slack). Finish Time, Most Successors, Minimum Slack).

Repository: github.com/prak05/AlignAI (Active repository with JSON-ingesting modularized Python engine).

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Program Outputs & Verification: GA & SA

Genetic Algorithm (GA) Execution Log:

- Permutation encoding with topological validation.
- Configuration: Population Size = 40, Generations = 100.

[SCREENSHOT PLACEHOLDER 1: GA Terminal Execution Log]

Simulated Annealing (SA) Execution Log:

- Adjacent swap neighborhood with precedence validation.
- Configuration: Iterations/Temp = 50.

[SCREENSHOT PLACEHOLDER 2: SA Terminal Execution Log]

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Program Outputs & Verification: BCO & PRH

Bee Colony Optimization (BCO) Execution Log:

- Setup: Food sources mapped to precedence-feasible schedules; nectar amount evaluated via reciprocal makespan.
- Configuration: Colony Size = 30, Employed = 15, Onlookers = 15, Max Iterations = 50.

```
[SCREENSHOT PLACEHOLDER 3: BCO
Terminal Execution Log]
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PRH Benchmark Comparison on Instance:

- Deterministic single-pass construction heuristics (LFT, MIS).

[SCREENSHOT PLACEHOLDER 4: PRH vs. Metaheuristic Gantt Comparison]

Algorithm	Method Category	Makespan	Runtime
PRH (LFT)	Priority Rule Heuristic	19 days	< 0.01s
BCO	Swarm Intelligence	17 days	0.42s
SA	Trajectory Local Search	17 days	0.28s
GA	Evolutionary Population	17 days	0.65s

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Project Conclusion

Groundwork Established:

- We have successfully identified the computational friction in traditional project scheduling and proposed an enhanced AI-driven system grounded in a thorough literature survey.
- The integration of **8 system improvements**—specifically Tabu Search (TS), Particle Swarm Optimization (PSO), and dynamic mid-search adaptability—establishes a robust theoretical foundation for superior scheduling.



Review 1 Deliverables Achieved:

- **30% Codebase Milestone:** Successfully developed, tested, and validated all 4 baseline algorithms (GA, SA, BCO, PRH) along with the Serial Schedule Generation Scheme (SSGS).
- **Empirical Validation:** Verified distinct runtime and makespan trade-offs across heuristics and metaheuristics, proving the necessity of an AI-driven automated selection model.
- **Phase 2 Preparation:** Established a clean, modular Python architecture ready for TS, PSO, feature extraction, and Graph Neural Network selector training.

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Academic References

[1] H. M. H. Saad et al., 'Quantum-Inspired Genetic Algorithm for Resource-Constrained Project Scheduling,' 2021.

[2] A. Corsini, S. Calderara, and M. Dell'Amico, 'Learning the Quality of Machine Permutations in Job Shop Scheduling,' 2022.

[3] L. Chen et al., 'APULSE: A Scalable Hybrid Algorithm for the RCSPP on Large-Scale Dense Graphs,' 2026.

[4] A. M. Al-Zoubi et al., 'Exploiting Ontology Recommendation Using Text Categorization Approach,' 2020.

[5] M. A. Al-Bakry et al., 'Ontology-Based Personalized Course Recommendation Framework,' 2019.

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THANK YOU

Open for any questions

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